Dr. S. AMUTHA
Assistant Professor

Contact

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Academic Qualifications: M.Sc./M.Phil./Ph.D./PGDCA.

Teaching Experience: 11 Years 6 Months

Research Experience: 9Years

Areas of Research

Graph Theory; Domination theory; Discrete Mathematics, Algorithemic Graph Theory.

Research Supervision / Guidance

Program of Study		Completed	Ongoing
Research	Ph.D.	1	1
	M.Phil.	30	2
Project	PG	71	14

Publications

- K.Suriya Prabha, **S.Amutha,** N.Anbazhagan and S.Sankara Gomathi**,** **"**Neighborhood outer Split Domination in Graphs", Journal of Discrete Mathematical sciences and Cryptography. Vol.22, No.5 , 2019, pp.787-799, ISSN: 0972-0529. DOI:10.1080/09720529.2019.1681695. [scopus].

- Sankara Gomathi. S , **Amutha. S** , Sridhar. G, Jayaprakasan.M. "Interpretation of Formal semantics from Hand Gesture to text using Proficient contour tracing technique", International Journal of Recent Technology & Engineering Volume 8, Issue 2S11, September, 2019, pp.2624-2629, ISSN 2277-3878, DOI:10.35940/ijrte.B1318.0982S1119 [scopus].

- Sankara Gomathi. S , **Amutha. S** , Jayaprakasan. "Personify Educational Assistance application for special children using deep learning" International journal of Innovative Technology and Exploring Engineering , Volume 8, Issue 10, August, 2019, pp.1609-1614, ISSN 2278-3075, DOI: 10.35940/ijitee.A1042.0881019.

- Sankara Gomathi. S , **Amutha. S** , Latha . A, Jayaprakasan. "Design of Power Optimization of Reversible Carry Select Adder using MPFA"", International Journal of Emerging Technology & Advanced Engineering Volume 8, Issue 10, October, 2018, pp.58-63, ISSN 2250-2459.

- K.Suriya Prabha and **S.Amutha, "**Split Domination number of a Congruent Dominating Graphs", International Journal of Pure and Applied

Mathematics. Vol.119, No.12 , 2018, pp.14633-14642, ISSN: 1314-3395. [IF:7.12].

- **S.Amutha "**Vertex odd and even mean labeling of a Regular graph", Journal of Advanced Research in Applied Sciences. Vol.5(3), pp.378-394, March 2018, DOI:16.10089.JARAS.2018.V513.453459-2464 .

- K.Suriya Prabha and **S.Amutha,** "Edge Mean Labeling of a Regular Graphs", International Journal of Mathematics Trends and Technology. Vol. 53, Number 4, ISSN No. 2231-5373 (2018) 24-33, [Seventh Sense Research Group].
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- N.Sridharan**, S.Amutha** and A.Ramesh Babu, " Bounds for Lamda Domination Number Gamma Lamda of G of a Graph", Journal of Computational Mathematica. Vol.1(2), pp.78-90, November 2017, doi.org/10.26524/cm18.

- N. Sridharan, **S. Amutha** and S.B. Rao, Induced Subgraphs of Gamma Graphs, Discrete Math. Algorithm. Appli. Vol. 5, Issue No. 3, 1350012 (2013) 1-5, DOI:10.1142/S1793830913500122 [World Scientific Publication].

- S. Amutha and N. Sridharan, " A note on sets V_t^-, V_t^0, V_t^+ of a simple graph G with $\delta(G) \geq 2$" , Journal of Pure and Applied Mathematics: Advances and Applications Vol. 9, Number 2, (2013) 69-79 [Scientific Advances Publishers].

Papers Published in Conference Proceedings and Books Chapter:

1. N. Sridharan and **S.Amutha , -** Characterization of Total very excellent trees, Mathematics and Computing, **Springer Proceedings** in Mathematics and Statistics, ISSN: 2194-1009, 2194-1017 (electrionic) ISBN 978-81-322-2451-8, 978-81-322-2452-5 (eBook) (2015), Vol 139, pp.265-275 DOI:10.1007/978-81-322-2452-5_18 (Springer India 2015) (Book Chapter 18).

2. S.Amutha and N. Sridharan, - γ_t - Graph of a Graph G, Lecture Notes Series, Number 7, Discrete Mathematics, Proc. ICDM 2006, ISBN: 978-81-902545-7-1 pp.255-262 [Ramanujan Mathematical Society]

3. K. Subramanian**, S.Amutha** and N. Sridharan, Gamma exchangeable graph

of a graph G, Proc. NCFOGTA'12 (2012) 88-94.

Poster & Paper Presentations in Conferences:

1. **Presented** a paper in an International Conference on Pure and Applied Mathematics on 15th February 2020 at Padmavani Arts and Science College for Women, Tamil Nadu, India.

2. **Presented** a paper in an International Conference on Algebra, Analysis and their Applications on 11th January 2020 at Madurai Kamaraj University, Madurai, Tamil Nadu, India.

3. **Presented** a paper in an International Conference on Mathematical Engineering Applications on 16th August 2019 at University of Malaysia, Kuala Lumpur.

4. **Presented** a paper in the National Conference on Mathematical Applications in Algebraic graph theory, Optimization technique and network theory (NCMA2018) on 2nd February 2018 at Mother Teresa Women's University, Kodaikanal, India.

5. **Presented** a paper in an International Conference on Mathematics and computing" held during 05-10 January 2015 at Haldia Institute of Technology, Haldia, West Bengal, India.

6. **Presented** a paper in "National Conference on Graph Theory Generalization and Applications" – NOGTGA- 2012 held during 20-21 August 2012 at Sastra Universitsy, Tanjavur.

7. **Presented** a paper in National Conference on "Fuzzy Optimization and Graph Theory with Applications" – NCFOGTA 12 held during 23-24 February 2012 at Mepco Schlenk Engineering College, Sivakasi.

8. **Presented** a paper in the National Conference on "Challenges in Mathematical Modeling" held on 22 December 2011 at Alagappa University, Ramanujan Centre for Higher Mathematics, Karaikudi.

9. **Presented** a paper in the Platinum Jubilee 75th Annual Conference of the Indian Mathematical Society (IMS) during December 27 – 30, 2009 at n-CARDMATH, Kalasalingam University, Krishnankoil.

10. **Presented** a paper in the National Conference on Recent Trends in Applicable Mathematics for Engineering and Technology on 2nd and 3rd April, 2009 at K.L.N. College of Engineering, Pottapalayam, Sivagangai Dist.

11. **Presented** a paper in the National Conference on "Algebra, Graph Theory and their Applications" organised by the Department of Mathematics, Manonmaniam Sundaranar University, Tirunelveli from 05.12.2007 to 07.12.2007.

12. **Presented** a paper in the International Conference on Discrete Mathematics held between December 15 – 18, 2006 at the Department of Mathematics, Indian Institute of Science, Bangalore.

13. **Presented** a paper in XXX Indian Social Science Congress, held between December 27 – 31, 2006 at Alagappa University, Karaikudi.

Funded Research Projects

Completed Projects

S. No	Agency	Period		Project Title	Budget (Rs. In lakhs)
		From	To		
1	UGC	24 Feb 2014	24 Feb 2016	A Study on various half-domination parameters of a graph	6,00000

Distinctive Achievements / Awards

- **Eminent Mathematician Award** from The International Multidisciplinary Research Foundation for the year 2019 at Malaysia.
- Received **Bright Educator Award** from International Institute of Organized Research for the year **2017**.

Events organized in leading roles

- Organizing Secretary of an International Webinar on Recent Advances in Graph Theory during May 29 - 30, 2020.

- Organizing Secretary of Workshop on MAPLE for Business Applications – Hands on Training on 26th February , 2019.

- Organizing Secretary of Workshop on Mathematical software for Business Applications – Hands on Training on 22nd February, 2019.

- Act as a Co-ordinator for the State Level Quiz Contest 2018" for PG Mathematics Students on 9th, March 2018 at Alagappa University, Karaikudi.

- Organizing Secretary of an International Conference on "Discrete and Computational Mathematics (ICDCM 2018)" under the financial support of

UGC-SAP(DRS-1) during 20-21, December , 2018 at Alagappa University, Karaikudi.

- Act as a Co-ordinator for the State Level Quiz Contest for PG Mathematics students held on 4th March, 2014 Alagappa University, Karaikudi.

- Joint Secretary of the organizing committee in the National Conference on Pure and Applied Mathematics (NCPAM16) held during 30-31 March, 2016 Alagappa University, Karaikudi

- Act as a Coordinator of the National Mathematics day on 22nd December 2016 at Alagappa University, Karaikudi.

- Act as a Co-ordinator for the State Level Quiz Contest for PG Mathematics students held on 4th March, 2014 Alagappa University, Karaikudi.

Events Participated

1. **Participated** in the Three-Day workshop on "Research Methods and LaTex" from 24th to 26th June, 2019 at Alagappa University, Karaikudi.

2. **Participated** in the one day workshop on "Design and Development of MOOCs for the Faculty Members of Alagappa University and Affiliated colleges" held on 8 –th November, 2018 at Alagappa University, Karaikudi.

3. **Participated** in the National level workshop on "Citizenship Skills for Pre-Service and In-Service Teachers" held on 10 –th February , 2017 at Alagappa University, Karaikudi.

4. **Participated** in the workshop on "R with Application to Quantitative Finance" held during August 26 – 28, 2016 at Birla Institute of Technology and Science Pilani, K K Birla Goa Campus, Goa (India).

5. **Participated** in the pre – conference workshop on "Algebraic and Applied Combinatorics" held during June 15 – 17, 2016 at Ramanujan Mathematical Society, National College, Tiruchirapalli, Tamil Nadu.

6. **Participated** in the National Level workshop on "Research methods and data analysis using SPSS" held on 18th March 2016 at Thanthai Roever College, Perambalur, Tamil Nadu.

7. **Participated** in the Two day National seminar on "Safety and security of women and girls in public spaces: Planning and designing mechanism" held on 4th – 5th December, 2015 at Alagappa University, Karaikudi , Tamil Nadu.

8. **Participated** in the Two day National Conference on " Impact of Electronic Resources on Teaching, Learning and Research: Issues and opportunities" held on 24th and 25th April, 2015 at Alagappa University, Karaikudi , Tamil Nadu.

9. **Participated** in an International Conference on "Mathematics and computing" held during 05-10 January 2015 at Haldia Institute of Technology, Haldia, West Bengal, India.

10. **Participated** in the National Conference on "Mathematical Modeling and Soft Computing" held on September 25th and 26th, 2014 at College of Engineering, Vatakara, Kerala.

11. **Participated** in the National Conference on "Recent Advances in Numerical Methods and its Applications" RANMA - 2014 held on January 27 - 28, 2014 at Pondicherry University, Puducherry.

12. **Participated** in the National Conference on "Frontiers In Analysis And Differential Equations" (NCFADE – 2012) held on December 19 - 20, 2012 at Bharathidasan University, Tiruchirappalli.

13. **Participated** in an International Conference on "Legacy Of Srinivasa Ramanujan" held on December 14 - 15, 2012 at Srinivasa Ramanujan Centre, Sastra University.

14. **Participated** in the National Conference on "Recent Trends In Mathematics And Its Applications" held on December 22, 2010 at Alagappa University, Karaikudi.

15. **Participated** in the National Seminar On "Analysis and Number Theory" held on 21 and 22, January 2010 at Alagappa University, Karaikudi.

16. **Participated** in the "One Day Seminar on Mathematics" held on 22.12.2009 at School Of Mathematics, Alagappa University, Karaikudi.

17. **Participated** in the "National Seminar on Applications of Algebra and Number Theory" held during December 4 – 5, 2009 at National College, Tiruchirapalli.

18. **Participated** in the "Instructional Lecture series in Algebra" held on 30.09.2009 at Alagappa University, Karaikudi.

19. **Presented** a paper entitled on "Tamizhaka Arivial pearavai onpatham karutharangam" between September 11 – 13, 2009 at Alagappa University, Karaikudi.

20. **Participated** in the National Seminar on "Fuzzy Mathematics and its Applications" on 9th March 2009 at Alagappa University, Karaikudi.

21. **Participated** in the "National Seminar on Differential Equations" on March 6 & 7 2009 at Alagappa University, Karaikudi.

22. **Participated** in an "International Conference on Stochastic Modelling" (ICSM09) during February 12 – 14, 2009 at Alagappa University, Karaikudi.

23. **Participated** in the Srinivasa Ramanujan Memorial Symposium on April 26, 2007 at SASTRA University, Srinivasa Ramanujan Centre, Kumbakonam, India.

24. **Participated** the lecture series on First INDO -US Lecture Series On "Discrete Mathematical Chemistry" between January 8 to 11, 2007 at PES Institute of Technology, Bengalooru, India.

25. **Participated** in the National Seminar on "FUZZY Mathematics and its Applications" at Department of Mathematics, Alagappa University, Karaikudi on 3rd and 4th March, 2006.

Other Training Programs

Orientation\Refresher courses\Faculty Development Programmes:

1. **Participated** in the **Two-week online Capacity Building Programme** through online mode Organized by Alagappa University under UGC STRIDE Component -1 Scheme from 12th June 2020 to 23rd June 2020.

2. **Participated** in One Week **Faculty Development Programme** on "Open Source Tools for Research" through online mode organized by Teaching Learning Centre, Ramanujan College, University of Delhi from June 08 - June 14, 2020 under HRDC.

3. **Participated** in One Week **Faculty Development Programme** on "Managing Online Classes and Co-Creating Moocs:2.0" through online mode organized by Teaching Learning Centre, Ramanujan College, University of Delhi from May 18 - June 03, 2020 under HRDC.

4. **Participated** in the **Refresher Course in Mathematics** obtained grade 'A' from July 13th to August 2nd , 2018 at UGC – HRDC, Barathidasan University, Trichirapalli.

5. **Participated** in the **Faculty Development(Enrichment) Program** ~ Accreditation Quality Enhancement ~ organized by Internal Quality

Assurance Cell(IQAC) from January 6, 2017 to January 12, 2017 at Alagappa University, Karaikudi.

6. **Participated** in the **Refresher Course in Mathematics** obtained grade 'A' from September 7th to September 27th , 2016 at UGC – HRDC, Pondicherry University, Pondicherry.

7. **Attended** the **"Faculty Development Programme on Entrepreneurship"** organized by the Entrepreneurship-cum Skill Development Centre from November 12, 2015 to November 26, 2015 at Alagappa University, Karaikudi.

8. **Participated** in the **Orientation Programme** obtained grade 'A' from May 30, 2013 to June 26, 2013 at Academic Staff College, Madurai Kamaraj University, Maduarai.

9. **Participated** in the **Orientation Programme** organized by the Curriculum Development Cell, Alagappa University, Karaikudi from 24 – 31st August, 2012 under the UGC initiative of National Mission on Teachers and Training.

10. **Participated** in the Summer school in Algebra and its Applications at Department of Mathematics, Alagappa University, Karaikudi from 22nd May to 9th June 2006.

Overseas Exposure / Visits

1. Malaysia
2. Thailand

Membership in

Professional Bodies

- Life member in Indian Mathematical Society(IMS).
- Life member in Kerala Mathematical Society.
- Life member in FATER Academy of India (Forum for Advanced Training Education and Research Academy of India).

- Life member in IAENG (International Association of Engineers).

Editorial Board

1. International Journal of Discrete Mathematics http://sciencepg.com/journal/editorialboard?journalid=605.
2. Special issue of Journal of Discrete Mathematical Sciences and Cryptography, volume 22, year 2019, pp. iii-iv, DOI:10.1080/09720529.2019.1695937

Academic Bodies (such as Board of Studies etc.,)

1. Board of studies member in Department of Mathematics, Tanjavur Tamil University, Tanjavur.
2. Board of studies member in Department of Mathematics, Alagappa University, Karaikudi.

Resource persons in various capacities

Plenary Talks/Invited Talks/Keynote Address :

1. Act as a **Chair person** in the "22nd FAI International Conference Jointly Organized by The American College , Madurai, India and FATER Academy of India on Mathematical, Computational Intelligence and Engineering Approaches to Healthcare, Business and Tourism Analytics 2020 (22nd FAI_ICMCIE 2020)" 21st, December, 2020.

2. Delivered a lecture on "**Modelling in Graphs**" at Seethalakshmi Achi College for women, Pallathur on 2nd, January 2020.

3. Delivered a talk in the State level seminar on Mathematical Analysis and Applications, Sivanthi Aditanar college, Pillayarpuram on 14th, October 2019.

4. Delivered an invited talk on "**Gamma Graphs and their Properties**" in the International Conference on Computational and Applicable Mathematics, The Quaide Milleth college for men, Chennai on 6th, September 2019.

5. Delivered an invited talk on "**Gamma Excellent Graphs**" at Institute of Mathematical Sciences, University of Malaya, Malaysia on 19th, August 2019.

6. Delivered an invited talk on "**Total very excellent caterpillers**" in the International Conference on Novel Trends in Pure and Applied Mathematics at University of Kerala, Thiruvananthapuram on 23rd January 2019.

7. Presented as a **Resource Speaker** on "**Domination in Graphs**" during 20th November 2017 to 3rd December 2017 the conduct of two weeks **faculty development programme(FDP)** on **"Mathematical Sciences Research Challenges in Engineering Applications"**at Sethu Institute of Technology, Pulloor, Kariapatti, Tamilnadu, India.

8. Delivered a **Plenary Talk** in an International Science Congress at Asian Institute of Technology, **Bangkok, Thailand** during October 02-06, 2017.

9. Act as a **Chair person** in the "International Conference on Mathematical Modelling and Computational Methods in Science and Engineering" during 20-22 February 2017 at Alagappa University, Karaikudi.

10. Delivered a **key-note address** in Two days State Level Science Symposium at Alagappa Chettiar College of Engineering and Technology, Karaikudi on 1st , March 2017.

11. Act as a **Chair person** in the "**International Conference on Discrete and Computational Mathematics"** held during February 16-18, 2017 at Gandhigram Rural Institute- Deemed University, Dindigul.

12. Delivered a **Plenary Lecture** on "**Graphs and its Applications**" in the National Seminar on Neoteric Tenor in Graph Theory at Dr.Umayal Ramanathan College for Women, Karaikudi on 15th February 2017.

13. Delivered a **key-note address** on "**Menger's theorem**" as a Chief Guest in Maths Association at APSA College, Tiruppattur on 19th September 2014.

Recent Publications

1. S. Amutha K. Suriya Prabha, N.Anbazhagan and P. Shanhi "Split domination number of divisible dominating graphs", Journal of Discrete Mathematical Sciences and Cryptography, ISSN 0972-0529 (Print), ISSN 2169-0065 (Online), DOI : 10.1080/09720529.2020.1744813 (Taylor and Francis) (Scopus) (Accepted).

2. K. Jeganathan, N. Anbazhagan, **S. Amutha**, V. Vinitha , "A queueing inventory system with threshold based priority service and a flexible server", *Journal of Statistics & Management Systems*, ISSN 0972-0510 (Print), ISSN 2169-0014 (Online), DOI: 10.1080/09720529.2020.1744811 (Taylor and Francis)(web of science) (Accepted).

3. N. Nithya, N. Anbazhagan, **S. Amutha**, K. Jeganathan and B. Koushick, " Working Vacation in Queueing – Stock system", Global and Stochastic Analysis, Vol. 8, No. 1, 2021, pp. 53-71.